LESSON PLAN

GAS TURBINE ENGINE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Fuel system, pressure control and flow control system, plunger types pump, gear type pump, various stages of fuel atomization and types of burners
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Fuel system, pressure control and flow control system, plunger types pump, gear type pump, various stages of fuel atomization and types of burners.
5. Specific Objectives	13.1.3.1 Trainee will be able to explain components and operation of basic engine fuel system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of fuel system schematic diagrams; identification of pump types and burner stages on wall charts; instructor supervision of diagram tracing and component labeling.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the fuel system in a gas turbine engine?
(b) Name two types of fuel pumps used in aircraft engines.
(c) Why is precise fuel atomization critical for engine combustion?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Fuel system components and operation • Pressure control and flow control mechanisms • Plunger type pump construction and operation • Gear type pump construction and operation	L & Ds	PPT, IB	15 min
(b)	• Stages of fuel atomization process • Types of fuel burners and their operation • Comparison of plunger and gear pump applications	L & Ds	PPT, PDF	20 min
(c)	• Summary of fuel system pressure regulation • Review of atomization efficiency factors • Integration of pumps and burners in engine cycle	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a plunger pump differs from a gear pump in terms of pressure generation.
(b) Describe the three main stages of fuel atomization and their purpose.
(c) Identify the key components involved in fuel flow control within the system.

12. Home Assignment and Follow Up	Prepare a brief note comparing the operation of plunger type and gear type pumps, highlighting their specific roles in the upcoming lesson on thrust augmentation (LP 11).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Thrust augmentation and Thrust augmentation by water injection and water methanol injection system After burning operation	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min